Lattice Yang–Mills theory at strong coupling IV: the mass gap is $4\log(1/\beta) + O(1)$

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d \geq 3$ and $0 < \beta < \beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II, and the finite-volume gaps of [F] on spatial lattices with all sides ≥ 4, are at least $4\log(1/q(\beta))$, with $q(\beta) = e^3 K(e^{4\beta} - 1)$ and $K = 4(2d-3)$. As $\beta \downarrow 0$ this is $4\log(1/\beta) - O(1)$. The leading coefficient 4 is the one predicted by the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, with predicted mass $-4\log u(\beta) + o(1)$, $u(\beta) \sim \beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta) - O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four (sides ≥ 4 are needed: with a side of length 3 the girth is three). In $d = 2$ the girth is the spatial size, which gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem. We also prove the matching upper bound: for an irreducible ρ of dimension N, $m \leq 4\log(1/u(\beta)) + D(\beta)$ with $u = c_\rho/(Nc_0)$ and an explicit $D(\beta) = O(\beta)$, on the whole range $0 < \beta < \beta_0(d)$ when $N \leq 3$ (then $D < 0.32$; numerically $D < 0.05$ for $U(1)$, $SU(2)$, $SU(3)$ in $d = 3, 4$). The proof uses a positivity property of the transfer matrix that isolates the minimal tube (the four sides of a unit cube between two plaquettes), an exact evaluation of that tube by character orthogonality, a lemma that every other term of the resulting expansion has at least five further plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\mathrm{Var}$ of the ground-state chain of Part II. Together: $4\log(1/q(\beta)) \leq m(\mu_\beta) \leq 4\log(1/u(\beta)) + D(\beta)$ for all $\beta < \beta_0(d)$, so $m(\mu_\beta) = 4\log(1/\beta) + O(1)$, which is what principle 3 ("the strong-coupling test") of the research program requires. The additive constants of the two bounds still differ (by about 35 for $SU(2)$ in $d = 4$); the difference comes from the lower bound.

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1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient for all $\beta < \beta_0(d)$ (Theorem 1.2) and, on the same range, the matching upper bound (Theorem 1.3); together, $m(\mu_\beta) = 4\log(1/\beta) + O(1)$.

Literature. The strong-coupling glueball spectrum was studied rigorously in the 1980s: by their abstracts, Schor [Sch83] proved the existence of glueballs in strongly coupled lattice gauge theories, and O'Carroll, Braga and Schor [OBS85] (in $2+1$ dimensions) and O'Carroll and Barbosa [OB85] (in $3+1$ dimensions) obtained isolated masses of the form $-4\log\beta + r(\beta)$. We have not yet read these papers (they are marked "verify"), and nothing below relies on them. What this article adds for the project is a short self-contained proof with explicit constants and range, for the infinite-volume theory of [II], and a covariance bound for all bounded local observables.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2)},$$

where $c(\beta) := (e^{2\theta(\beta)}(1 + \frac{1}{K}))^{2(d-1)}$.

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Theorem 1.3 (Upper bound; the gap is $4\log(1/\beta)+O(1)$)**.** *Let* G, ρ *be as in Theorem 1.2 (*G *metrisable),* $d \geq 3$*, and let* ρ *be irreducible and non-trivial, of dimension* $N \leq 3$ *(for instance* $U(1)$*,* $SU(2)$*,* $SU(3)$*); put* $\iota := 1$ *if* $\rho \cong \bar{\rho}$ *and* $\iota := 0$ *otherwise, and* $u(\beta) := c_\rho(\beta)/(Nc_0(\beta))$*. There is an explicit function* $D(\beta)$ *(Theorem 5.6) with* $0 < D(\beta) < 0.32$ *and* $D(\beta) = O(\beta)$ *such that for every* $0 < \beta < \beta_0(d)$

$$4\log\frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4\log\frac{1}{u(\beta)} + D(\beta);$$

the upper bound also holds for $m_{\Lambda_s}(\beta)$ *on every periodic spatial lattice with all sides* ≥ 3*, and both bounds on every periodic spatial lattice with all sides* ≥ 4*. Hence*

$$-4\log(4e^3K) - O(\beta) \;\leq\; m(\mu_\beta) - 4\log\frac{1}{\beta} \;\leq\; 4\log\frac{2N^2}{1+\iota} + O(\beta) \qquad (0 < \beta < \beta_0(d)).$$

For every N*, the upper bound holds for all sufficiently small* β*, so* $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ *as* $\beta \downarrow 0$.

Here $\mathrm{Cov}(A, B) := \mu(AB) - \mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) follows from Theorem 4.1 by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)], and (ii) from (iii) by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii).

Remark 1.4 (Comparison). For $d = 4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta) - 2.89 + O(\beta^2)$; the bound here is $4\log(1/\beta) - 29.5 + O(\beta)$. In the common range of validity $\beta < \beta_0(4) \approx 1.54 \cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (5.890 and 5.591 at $\beta_0(4)$), and the present one is larger for $\beta < 1.39 \cdot 10^{-4}$, by an additive amount that tends to infinity as $\beta \to 0$ (the ratio of the two bounds increases to 4; it is 1.92 at $\beta = 10^{-5}$). These values were computed with `python3`. The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant -29.5 is far from the predicted $+4\log 4$ for $SU(2)$ (from $u \approx \beta/4$).

Strategy. (1) Let U and U' be two independent copies of the field, each with law μ. Then $2\,\mathrm{Cov}_\mu(A, B)$ is the integral of $(A - A')(B - B')$ against $\mu \otimes \mu$. The duplicated Wilson weight is $\prod_p(1 + f_\beta(U_p))(1 + f_\beta(U'_p)) = \prod_p(1 + g_p)$ with $|g_p| \leq e^{4\beta} - 1$, so the loop expansion of [III] applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U \leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.2), hence at least $4\times$(number of layers) plaquettes (Corollary 3.4). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III]. (5) The upper bound (Section 5) uses positivity instead of cancellation. Splitting the one-link operator of the transfer matrix into its trivial and non-trivial parts writes the covariance of two plaquettes at times 0 and 1 as a sum of non-negative terms; one of them is the expectation of a local function in which the four sides of the unit cube between the plaquettes carry non-trivial characters. Its loop expansion is evaluated exactly on the cube by character orthogonality, every other term has at least five further plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\,\mathrm{Var}$ gives $m \leq 4\log(1/u) + O(\beta)$ on the whole range $\beta < \beta_0(d)$.

2 The duplicated system

Throughout this section and the next, $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda \geq 3$, with the conventions of [F, III]: $\mathcal{P} = \mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta = w_\beta/c_0(\beta) - 1$, and for $\mathcal{Q} \subseteq \mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q}) = \int \prod_{p\in\mathcal{Q}}(1 + f_\beta(U_p))\, dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\, dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U, U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p\in X} g_p\, dU\, dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \leq \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p\in\mathcal{Q}}(1 + g_p)\, dU\, dU' = \Xi(\mathcal{Q})^2$*; and* $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ *has density* $\prod_{p\in\mathcal{P}}(1 + g_p)/\Xi(\mathcal{P})^2$ *with respect to* $dU\, dU'$.

(c) Let $X \subseteq \mathcal{P}$*, let* ℓ *be a link lying in exactly one plaquette of* X*, and let* H *be bounded measurable and independent of* (U_ℓ, U'_ℓ)*. Then* $\iint H \prod_{p\in X} g_p\, dU\, dU' = 0$.

Proof. (a) By [III, Lemma `al:lem:facts`](a),(b), $e^{-\beta} \leq w_\beta \leq e^\beta$ and $1 \leq c_0 \leq e^\beta$, so $1 + f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^\beta]$ and $(1 + f_\beta(U_p))(1 + f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta} - 1, e^{2\beta} - 1]$; both $1 - e^{-4\beta}$ and $e^{2\beta} - 1$ are at most $e^{4\beta} - 1$. (b) The integrand is $\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p)) \cdot \prod_{p\in\mathcal{Q}}(1+f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U)W_\beta(U') = c_0^{2|\mathcal{P}|} \prod_p (1 + g_p)$ and $Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm 1}B$ and $U'_p = A'U'^{\pm 1}_\ell B'$ with A, B, A', B' independent of (U_ℓ, U'_ℓ), and no other factor of the integrand depends on (U_ℓ, U'_ℓ). By Fubini it suffices that $\iint g_p\, dU_\ell\, dU'_\ell = \int (1 + f_\beta(AU^{\pm 1}B))\, dU \cdot \int (1 + f_\beta(A'U^{\pm 1}B'))\, dU - 1 = 1 \cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p) \cap S \neq \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X) \cup S) \neq \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \leq J_S$ and $|\overline{X}| \leq (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let* $S \subseteq E(\Lambda)$ *and let* F *be bounded, measurable, and dependent only on the variables* (U_ℓ, U'_ℓ)*,* $\ell \in S$*. Then*

$$\int F\, d(\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}) = \sum_{X\in\mathcal{X}_S} \Phi_F(X) \Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U'_ℓ), dU by $dU\, dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\, d(\mu \otimes \mu) = \sum_{T\subseteq\mathcal{P}} \iint F \prod_{p\in T} g_p\, dU\, dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P} \setminus \overline{X}$. For such a pair, $F \prod_{p\in X} g_p$ depends only on the variables of the links in $S \cup E(X)$ and $\prod_{p\in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y\subseteq\mathcal{P}\setminus\overline{X}} \iint \prod_{p\in Y} g_p = \iint \prod_{p\in\mathcal{P}\setminus\overline{X}}(1 + g_p) = \Xi(\mathcal{P} \setminus \overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B, put $S := S_A \cup S_B$ and $F(U,U') := (A(U)-A(U'))(B(U)-B(U'))$. Then:*

(i) $\int F\, d(\mu\otimes\mu) = 2\,\mathrm{Cov}_\mu(A,B)$ *for every probability measure μ on Ω_Λ, and $\|F\|_\infty \le 4\|A\|_\infty\|B\|_\infty$;*

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A-A')(B-B')\,d(\mu\otimes\mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X\setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1)\cap S_B = \emptyset$, and likewise $E(X_2)\cap S_A = \emptyset$; and $E(X_1)\cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A \cup E(X_1)$ and $S_B \cup E(X_2)$ are disjoint, and since $dU\,dU'$ is a product measure,

$$\Phi_F(X) = \iint (A-A') \prod_{p\in X_1} g_p \;\cdot\; \iint (B-B') \prod_{p\in X_2} g_p.$$

The exchange $(U,U') \mapsto (U',U)$ preserves $dU\,dU'$ and every g_p (which is symmetric in U_p, U_p'), and changes the sign of $A-A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F\prod_{p'\in X\setminus\{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x,0)$ is *temporal of layer x_0* (it joins times x_0 and x_0+1); a plaquette $(x,0i)$, $1\le i\le d-1$, is *temporal of layer x_0*. For $Y\subseteq\mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one). Let $\Lambda_s := \prod_{i=1}^{d-1}\mathbb{Z}/L_i\mathbb{Z}$ be the spatial torus and g the *girth* of its nearest-neighbour graph (the length of a shortest cycle).

Lemma 3.1 (Girth)**.** *Let all $L_i \ge 3$. If $d\ge 3$, then $g = 4$ when all $L_i \ge 4$, and $g = 3$ when some $L_i = 3$. If $d = 2$, then $g = L_1$.*

Proof. Since all $L_i \ge 3$ the graph is simple. A cycle is a closed walk without backtracking, i.e. a sum of unit vectors $\pm e_i$ equal to 0 in Λ_s with no two consecutive steps opposite. For $d = 2$ the graph is the cycle $\mathbb{Z}/L_1\mathbb{Z}$ itself. For $d\ge 3$ a unit square is a cycle of length 4; a cycle of length 3 needs $\pm e_i \pm e_j \pm e_k = 0$ with no two opposite consecutive steps, which forces $i=j=k$ and $3e_i = 0$, i.e. $L_i = 3$; and conversely $L_i = 3$ gives the cycle $0, e_i, 2e_i$. □

Lemma 3.2 (Plaquettes per layer)**.** *Let all $L_i \ge 3$ $(1\le i\le d-1)$. Let $S\subseteq E(\Lambda)$ and $X\subseteq\mathcal{P}$ be such that every link of $E(X)\setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \ne \emptyset$. Then $|Y_s| \ge g$. In particular $|Y_s| \ge 4$ if $d\ge 3$ and all $L_i \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s,\vec{x}),0)$, $\vec{x}\in\Lambda_s$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s,\vec{x}),0i)$ and $((s,\vec{x}-e_i),0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s,\vec{x}),0i)\in Y_s$, each joining $\vec{x}$ and $\vec{x}+e_i$ (the positions of its two temporal links). Since $L_i\ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a subgraph of the nearest-neighbour graph of Λ_s. If $\vec{x}$ is an endpoint of an edge,

then $\ell_{\vec{x}} \in E(Y) \setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \neq \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats), which has at least g edges; they are distinct plaquettes of Y_s. The last statement is Lemma 3.1. □

Lemma 3.3 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \neq s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1, \dots, s'\}$ and $I_2 := \{s'+1, \dots, s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s, s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.4 (Size of connecting components)**.** *Let all $L_\lambda \geq 3$. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \leq a_2 < b_1 \leq b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1, \dots, a_2\}$ and those of S_B in $\{b_1, \dots, b_2\}$ (modulo L_0). Put*

$$\tau_{\text{in}} := b_1 - a_2, \qquad \tau_{\text{out}} := L_0 - (b_2 - a_1), \qquad \tau := \min(\tau_{\text{in}}, \tau_{\text{out}}) \quad (\geq 1).$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq g\,\tau$. More precisely, with the inner layers $\{a_2, \dots, b_1 - 1\}$ and the outer layers $\{b_2, \dots, a_1 - 1 + L_0\}$, Y contains at least g temporal plaquettes in every inner layer, or at least g temporal plaquettes in every outer layer.

Proof. Call the layers $s \in \{a_2, \dots, b_1 - 1\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2, \dots, a_1 - 1 + L_0\}$ outer (τ_{out} of them); as integers, $s < s' < s + L_0$ for every inner s and outer s', so inner and outer layers are distinct. *These layers are free.* A temporal link of an inner layer s joins s and $s+1$, where $s+1 \in \{a_2+1, \dots, b_1\}$ is not in $\{a_1, \dots, a_2\}$ and $s \in \{a_2, \dots, b_1 - 1\}$ is not in $\{b_1, \dots, b_2\}$ modulo L_0 (because $b_2 - a_1 < L_0$); so it belongs neither to S_A nor to S_B. The same argument applies to outer layers. *Times of the plaquettes of Y at the supports.* Let $p \in Y \cap \mathcal{P}_{S_A}$, containing a link $\ell \in S_A$. If p is spatial, $T(p) = \{r\}$ with $r \in \{a_1, \dots, a_2\}$; if p is temporal of layer r, then ℓ is either its temporal link of layer r or a spatial link at time r or $r+1$, so $r \in \{a_1 - 1, \dots, a_2\}$. Now let s be inner and s' outer with $Y_s = Y_{s'} = \emptyset$, and I_1, I_2 as in Lemma 3.3. Then $T(p) \subseteq I_2 = \{s'+1, \dots, s\}$: the times $a_1, \dots, a_2$ lie in I_2 because $s' + 1 \leq a_1 + L_0$ and $s \geq a_2$; the time $a_2 + 1$ occurs in $T(p)$ only if p is temporal of layer a_2, which is excluded when $s = a_2$ (as $Y_s = \emptyset$) and lies in I_2 when $s \geq a_2 + 1$; similarly the time $a_1 - 1$ occurs only if p is temporal of layer $a_1 - 1$, which is excluded when $s' \equiv a_1 - 1$ and lies in I_2 otherwise. In the same way $T(p) \subseteq I_1 = \{s+1, \dots, s'\}$ for every $p \in Y \cap \mathcal{P}_{S_B}$. *Conclusion.* If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.3 puts all of Y into one of the disjoint classes I_1, I_2, which contradicts the previous paragraph, since Y meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.2 (with $S = S_A \cup S_B$) gives at least g plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \geq 2$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \geq 3$; let g be the girth of Lemma 3.1. Let A, B be bounded*

measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.4, with $\tau = \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta) < 1$ (Lemma 4.2) and $c = c(\beta)$ as in Theorem 1.2,

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)| \le \frac{2\|A\|_\infty \|B\|_\infty}{1-q}\, c^{|S_A|+|S_B|}\, q^{g\tau}.$$

Proof. If S_A or S_B is empty, A or B is constant and the covariance vanishes; so let both be non-empty. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\,\mathrm{Cov}(A,B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X)\setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.4 these have $|X| \ge n_0 := g\tau$. For each of them: $|\Phi_F(X)| \le 4\|A\|_\infty\|B\|_\infty\,\epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2 \le e^{2\theta|\overline{X}|} \le e^{2\theta(K+1)|X|} e^{2\theta J_S} \le e^{2|X|} e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S+Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \le J_S$ and the adjacency bound K; it requires $J_S \ge 1$, i.e. $S \ne \emptyset$). With $\binom{m}{n}t^n \le (1+t)^m$ at $t = 1/K$, $\binom{J_S+Kn}{n} \le (1+\frac{1}{K})^{J_S}(1+\frac{1}{K})^{Kn}K^n \le (1+\frac{1}{K})^{J_S}(eK)^n$. Altogether

$$2|\,\mathrm{Cov}(A,B)| \le 4\|A\|_\infty\|B\|_\infty\, e^{2\theta J_S}\Big(1+\frac{1}{K}\Big)^{J_S} \sum_{n\ge n_0} \big(eK\cdot e^2\epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-q}\, q^{n_0},$$

since $e^3K\epsilon(2\beta) = q$ and $(e^{2\theta}(1+\frac{1}{K}))^{J_S} = c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d \ge 2$ and $0 < \beta < \beta_0(d)$: $q(\beta) < \frac{e(2+\epsilon_0)}{K+2}$, which is < 0.909 for $d = 2$ and < 0.389 for $d \ge 3$; and $4\log\frac{1}{q(\beta)} = 4\log\frac{1}{\beta} - 4\log(4e^3K) + O(\beta)$ as $\beta \downarrow 0$. Moreover $\beta_0(d) < \beta_*(d) = \operatorname{artanh}\frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta}-1 < \epsilon_0$ (which is the condition $\beta < \beta_0(d)$), $e^{4\beta}-1 = \epsilon(2+\epsilon) < \epsilon_0(2+\epsilon_0)$ and $e^3K\epsilon_0 = \frac{e}{K+2}$; for $d = 2$, $K = 4$ and $\frac{e(2+\epsilon_0)}{6} < 0.909$; for $d \ge 3$, $K \ge 12$ and $\frac{e(2+\epsilon_0)}{14} < 0.389$. The asymptotics follow from $e^{4\beta}-1 = 4\beta+O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta < \beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde{\mu}_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A \cup S_B$, the functions $A\circ\iota$, $B\circ\iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i, b_i, $g = 4$, $\tau_{\text{in}} = b_1 - a_2$ and $\tau_{\text{out}} \to \infty$. Theorem 4.1 bounds $\mathrm{Cov}_{\tilde{\mu}_{\Lambda_k,\beta}}(A,B)$ uniformly in k, and the left side converges to $\mathrm{Cov}_{\mu_\beta}(A,B)$.

(ii) Let $F \in \mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0, n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F, 0]$ and S_tF on links with endpoint times in $[t, t+n_F]$, with the same number of links. For $t \ge 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_tF) = |\mu_\beta(F)|^2$),

$$\big|\mu_\beta(\Theta F\cdot S_tF) - |\mu_\beta(F)|^2\big| = \big|\mathrm{Cov}_{\mu_\beta}(\Theta F, S_tF)\big| \le C_F\, q^{4t} \qquad (t \ge 1),$$

with C_F independent of t (here $b_1 - a_2 = t$); for $t = 0$ the left side is at most $2\|F\|_\infty^2$. So the hypothesis of [II, Theorem `iv:thm:OS`](iii) holds with $m = 4\log(1/q)$ and the constant $\max(C_F, 2\|F\|_\infty^2)$, and $m(\mu_\beta) \geq 4\log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s = G^{E(\Lambda_s)}$ and $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \geq 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.4 take $a_1 = a_2 = 0$, $b_1 = b_2 = t$; for $1 \leq t \leq L_0/2$, $\tau = \min(t, L_0 - t) = t$, and $g = 4$ since all $L_i \geq 4$. Theorem 4.1 gives $|\langle \overline{A}(t)A(0)\rangle - \langle \overline{A}(t)\rangle\langle A(0)\rangle| \leq C_A q^{4t}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0; for $t = 0$ the left side is at most $2\|A\|_\infty^2$. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0 \to \infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta) \geq 4\log(1/q)$. □

5 The matching upper bound

Throughout this section $d \geq 3$, G is metrisable (as in Theorem 1.2), and ρ is irreducible and non-trivial (automatic for an irreducible faithful ρ when $G \neq \{1\}$); $N = d_\rho$, $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, $\tau_0 := \frac{1+\iota}{2}$, and

$$u_r(\beta) := \frac{c_r(\beta)}{d_r\, c_0(\beta)} \quad (r \text{ irreducible}), \qquad u(\beta) := u_\rho(\beta) = \frac{c_\rho(\beta)}{N c_0(\beta)},$$

$$\varphi_\beta := 1 - \frac{c_0(\beta)}{w_\beta} = \frac{f_\beta}{1 + f_\beta}.$$

Since w_β is real and invariant under inversion, $c_r = \int \overline{\chi_r}\, w_\beta\, dU = \int \chi_r\, w_\beta\, dU = c_{\bar{r}}$ is real (and $c_r > 0$ by [F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]); so $u_{\bar{r}} = u_r > 0$, and $f_\beta = \sum_{r \neq 0} d_r u_r \chi_r$, absolutely and uniformly convergent ([F, Corollary `cor:uniform`]). Fix two spatial directions, labelled 1, 2. On a periodic lattice Λ (all $L_\lambda \geq 3$) let $p_0 := (0, 12)$ be the spatial plaquette at the origin and time 0, $p_1 := (e_0, 12)$ its translate at time 1, Q the unit cube with bottom p_0 and top p_1, $\mathcal{Q}$ the set of its six faces, $X_0 \subseteq \mathcal{Q}$ its four sides (the temporal plaquettes $(y, 0i)$ of layer 0 with $(y, i) \in E(p_0)$), $S_Q := E(Q)$ its twelve edges and $S_A := E(p_0)$; and

$$A := \operatorname{Re} \chi_\rho(U_{p_0}), \qquad A_1 := \operatorname{Re} \chi_\rho(U_{p_1}).$$

The same notation is used on $\mathbb{Z}^d$.

The proof has three steps. (a) A positivity property of the transfer matrix (Lemma 5.2) bounds $\operatorname{Cov}(A, A_1)$ from below by the expectation of one function of the twelve edges of Q, in which the four sides carry the factor φ_β, i.e. are forced to carry non-trivial characters. (b) The loop expansion of [III], applied to this local expectation, gives an exact integral over the cube, evaluated by character orthogonality (Lemma 5.1), plus terms with at least five further plaquettes (Lemma 5.4). (c) The spectral inequality $\operatorname{Cov} \leq e^{-m} \operatorname{Var}$. Unlike the duplicated expansion of Section 2, nothing here compares the leading term with the other surfaces connecting p_0 and p_1: the remainder is relative to the minimal tube itself. This is why the result holds on the whole range $\beta < \beta_0(d)$.

Lemma 5.1 (Character computations)**.** *For every $\beta > 0$:*

(i) $\int A\, dU = 0$ *and* $\int A^2\, dU = \tau_0 = \frac{1+\iota}{2}$.

(ii) Let a_0, a_1 be real continuous class functions on G, invariant under $U \mapsto U^{-1}$, with $\sum_r |\hat{a}_{k,r}| d_r < \infty$, where $\hat{a}_{k,r} := \int a_k \overline{\chi_r}\, dU$. Then $\hat{a}_{k,r} = \hat{a}_{k,\bar{r}} \in \mathbb{R}$ and

$$\int a_0(U_{p_0})\, a_1(U_{p_1}) \prod_{s \in X_0} f_\beta(U_s)\, dU = \sum_{r \neq 0} \hat{a}_{0,r}\, \hat{a}_{1,r}\, u_r^4.$$

In particular $\int A A_1 \prod_{s \in X_0} f_\beta(U_s)\, dU = \tau_0 u^4$.

(iii) $\frac{(1+\iota)\beta e^{-\beta}}{2N^2} \le u(\beta) \le \frac{(1+\iota)\beta+\beta^2 e^{\beta}}{2N^2}$.

(iv) $\int |\varphi_\beta|\, dU \le e^{3\beta}\beta\sqrt{\tau_0}/N \le e^{4\beta} N\, u(\beta)/\sqrt{\tau_0}$.

Proof. (i) $\int \chi_\rho = 0$ since ρ is irreducible and non-trivial. With $\operatorname{Re}\chi = \frac{1}{2}(\chi+\bar\chi)$, $\int(\operatorname{Re}\chi_\rho)^2 = \frac{1}{4}(\int \chi_\rho^2 + 2\int|\chi_\rho|^2 + \int \overline{\chi_\rho}^{\,2})$. By Schur orthogonality [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii), $\int|\chi_\rho|^2 = 1$ and, since $\chi_{\bar\rho} = \overline{\chi_\rho}$, $\int \chi_\rho^2 = \int \chi_\rho\overline{\chi_{\bar\rho}} = \iota$; likewise $\int \overline{\chi_\rho}^{\,2} = \iota$.

(ii) Substituting $U \mapsto U^{-1}$ (Haar measure is inversion invariant) and using $a_k(U^{-1}) = a_k(U)$, $\chi_r(U^{-1}) = \overline{\chi_r(U)}$: $\hat a_{k,r} = \int a_k \chi_r\, dU$, which is $\overline{\hat a_{k,r}}$ (a_k is real) and equals $\int a_k\overline{\chi_{\bar r}} = \hat a_{k,\bar r}$. Orient the six faces of Q as its boundary. The proof of [F, Proposition 'Cube activity' (`prop:cube`)] establishes, in its line "Altogether $\int\prod_p \chi_{r_p}(U_p)\, dU = \delta_{\text{all } r_p = r}\, d_r^{-4}$" (whose derivation does not use $r_p \neq 0$): for irreducible representations attached to the six faces, $\int\prod_{\text{faces}}\chi_{r_{\text{face}}}(U_{\text{face}})\, dU$ equals d_r^{-4} if all six are equivalent to the same r, and 0 otherwise. Each boundary-oriented face holonomy is conjugate to U_p or U_p^{-1}, and a_0, a_1, f_β are class functions invariant under inversion, so they are the same functions of the boundary-oriented holonomies. Insert $a_k = \sum_r \hat a_{k,r}\chi_r$ for the bottom and the top and $f_\beta = \sum_{r\neq 0} d_r u_r \chi_r$ for the four sides. The resulting six-fold series converges absolutely, because $|\int\prod\chi_{r_{\text{face}}}| \le \prod d_{r_{\text{face}}}$, $\sum_r |\hat a_{k,r}| d_r < \infty$ and $\sum_r d_r^2 u_r = \sum_r d_r c_r/c_0 = e^\beta/c_0$ ([F, eq. `eq:trace`]); so it may be integrated term by term. Only the terms in which all six faces carry the same r survive, and $r \neq 0$ because f_β has no trivial component; such a term contributes $\hat a_{0,r}\hat a_{1,r}(d_r u_r)^4 d_r^{-4}$. For $a_0 = a_1 = \operatorname{Re}\chi_\rho$: if $\iota = 1$, χ_ρ is real, $\hat a_{k,\rho} = 1$, and the sum is u^4; if $\iota = 0$, $\hat a_{k,\rho} = \hat a_{k,\bar\rho} = \frac{1}{2}$ and the sum is $2\cdot\frac{1}{4}u^4$. In both cases it is $\tau_0 u^4$.

(iii) By the proof of [F, Lemma `lem:cr`], $c_\rho = \sum_{n,m\ge 0}\frac{t^{n+m}}{n!\,m!} m_\rho(n,m)$ with $t = \beta/(2N)$ and $m_\rho(n,m)$ the multiplicity of ρ in $\rho^{\otimes n}\otimes\bar\rho^{\otimes m}$. Here $m_\rho(0,0) = 0$, $m_\rho(1,0) = 1$, $m_\rho(0,1) = \iota$, and $m_\rho(n,m) \le N^{n+m-1}$ (dimension count). Hence $(1+\iota)t \le c_\rho \le (1+\iota)t + N^{-1}(e^{2Nt} - 1 - 2Nt) \le \frac{(1+\iota)\beta}{2N} + \frac{\beta^2 e^\beta}{2N}$, using $e^y - 1 - y \le \frac{y^2}{2}e^y$ with $y = 2Nt = \beta$. Divide by Nc_0 and use $1 \le c_0 \le e^\beta$ ([III, Lemma `al:lem:facts`](b)).

(iv) As $1 + f_\beta = w_\beta/c_0 \ge e^{-2\beta}$ ([III, Lemma `al:lem:facts`](a),(b)), $|\varphi_\beta| \le e^{2\beta}|f_\beta|$, and by the Cauchy–Schwarz inequality $\int|\varphi_\beta| \le e^{2\beta}(\int f_\beta^2)^{1/2}$. Now $\int f_\beta^2 = c_0^{-2}\int(w_\beta - c_0)^2 \le \int(w_\beta - 1)^2$, because $c_0 \ge 1$ and $c = c_0 = \int w_\beta$ minimises $\int(w_\beta - c)^2$. With $y := \frac{\beta}{N}\operatorname{Re}\chi_\rho$ we have $|y| \le \beta$, $|e^y - 1| \le |y|e^{|y|}$, and so $\int(w_\beta - 1)^2 \le e^{2\beta}\int y^2 = e^{2\beta}\beta^2\tau_0/N^2$ by (i). This gives the first inequality; the second follows from $\beta \le e^\beta N^2 u/\tau_0$, which is (iii). □

Lemma 5.2 (Positivity)**.** *Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z})\times\Lambda_s$ be periodic with all sides ≥ 3, $\beta > 0$, B a real bounded measurable function on Ω_s, $b \in \mathbb{R}$, and $E_0 \subseteq E_s$ a set of spatial links; for $\ell \in E_s$ let s_ℓ be the temporal plaquette of layer 0 (between the slices 0 and 1) based on ℓ. Then, with $\Delta_B := (B(0) - b)(B(1) - b)$,*

$$\langle\Delta_B\rangle_{\Lambda,\beta} \;\ge\; \Big\langle \Delta_B \prod_{\ell\in E_0}\varphi_\beta(U_{s_\ell})\Big\rangle_{\Lambda,\beta}.$$

In particular, if $b = \langle B(0)\rangle_{\Lambda,\beta}$, which equals $\langle B(1)\rangle_{\Lambda,\beta}$ by translation invariance, the left side is $\operatorname{Cov}_{\Lambda,\beta}(B(0), B(1))$.

Proof. We use the transfer matrix $\hat T = VK_1PV$ of [F, Definition `def:T`] on $L^2(\Omega_s)$. By [F, Lemma 'The one-link operator' (`lem:kop`)], the one-link operator is $k_\beta = c_0\Pi_0 + k'_\beta$ with $k'_\beta := \sum_{r\neq 0}(c_r/d_r)\Pi_r$; both summands are positive operators ($c_r > 0$), with kernels c_0 and $w_\beta(UV^{-1}) - c_0$. For $F \subseteq E_0$ let $K_1^{(F)}$ be the tensor product over $\ell \in E_s$ of k'_β ($\ell \in F$), $c_0\Pi_0$ ($\ell \in E_0 \setminus F$) and k_β ($\ell \notin E_0$). Expanding $\bigotimes_{\ell\in E_0}(c_0\Pi_0 + k'_\beta)$ gives $K_1 = \sum_{F\subseteq E_0} K_1^{(F)}$. Each $K_1^{(F)}$ is positive, as a tensor product of positive operators, and its kernel is a product over the links of class functions of $U_\ell U_\ell'^{-1}$; so, as in the proof of [F, Lemma `lem:K`](i), it commutes

with every R_g and hence with P. Put $\hat{T}_F := VK_1^{(F)}PV = P(VK_1^{(F)}V)P$. Then $\hat{T}_F \geq 0$ and $\hat{T} = \sum_F \hat{T}_F$, so $0 \leq \hat{T}_F \leq \hat{T}$ and $\hat{T}_F$ is trace class.

Let $\tilde{B}$ be multiplication by $B - b$. By [F, Theorem 'Transfer matrix formalism' (`thm:tm`)],

$$\langle \Delta_B \rangle = \frac{\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}\tilde{B})}{\operatorname{Tr}\hat{T}^{L_0}} = \sum_{F \subseteq E_0} \frac{\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}_F\tilde{B})}{\operatorname{Tr}\hat{T}^{L_0}}.$$

Every term is non-negative: by cyclicity of the trace ($\hat{T}_F$ is trace class, the other factors are bounded), $\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}_F\tilde{B}) = \operatorname{Tr}(\hat{T}_F^{1/2}\,\tilde{B}\hat{T}^{L_0-1}\tilde{B}\,\hat{T}_F^{1/2})$, the trace of a positive operator. Finally, the term $F = E_0$ is the right side of the claim. Indeed, the proof of [F, Theorem `thm:tm`] applies verbatim when, in layer 0, the weights $w_\beta(U_{s_\ell})$, $\ell \in E_0$, are replaced by $w_\beta(U_{s_\ell})\varphi_\beta(U_{s_\ell}) = w_\beta(U_{s_\ell}) - c_0$, again a class function invariant under inversion: integrating out the temporal links of layer 0 then gives, by the computation in the proof of [F, Lemma `lem:K`](ii), the kernel of $K_1^{(E_0)}P$ instead of that of $K = K_1P$, i.e. $\hat{T}_{E_0}$ instead of $\hat{T}$ between the slices 0 and 1. □

Remark 5.3. Lemma 5.2 is a form of reflection positivity for the reflection through the plane $x_0 = \frac{1}{2}$, which bisects links ([F, Remark 'Other reflections'](ii)), applied to each piece of a decomposition of the weights of the crossing plaquettes into positive-definite parts. It holds for every $\beta > 0$. Its point is that it discards, with the right sign, every contribution to the covariance except the one in which the four sides of Q carry non-trivial characters.

Lemma 5.4 (Five plaquettes)**.** *Let Λ be periodic with all sides ≥ 3, or $\Lambda = \mathbb{Z}^d$. Let $B \subseteq E(\Lambda)$, and let $Y \neq \emptyset$ be a finite set of plaquettes such that every plaquette of Y contains at most one link of B and every link of $E(Y) \setminus B$ lies in at least two plaquettes of Y. Then $|Y| \geq 5$.*

Proof. Two facts. (a) Distinct plaquettes share at most one link. (b) For a link m and a vertex $v \notin m$, at most one plaquette contains both. For (a): two links of a plaquette either meet at a corner, and two perpendicular links at a common vertex lie in exactly one plaquette; or they are opposite, (z, μ) and $(z + e_\nu, \mu)$, and then the plaquette is $(z, \mu\nu)$, determined because $z + e_\nu \neq z - e_\nu$ when $L_\nu \geq 3$. For (b): with $m = (z, \mu)$, the plaquettes containing m are $(z, \mu\nu)$ and $(z - e_\nu, \mu\nu)$, $\nu \neq \mu$; their vertices off m are $z \pm e_\nu$, $z + e_\mu \pm e_\nu$, and these $4(d-1)$ points are distinct when all sides are ≥ 3, so v determines the plaquette.

Let $q \in Y$. It has at least three links outside B, each of which lies in a further plaquette of Y, and by (a) these further plaquettes are distinct; so $|Y| \geq 4$. Suppose $|Y| = 4$. Applying this to every plaquette of Y: each has exactly three links outside B (four would need four further plaquettes) and exactly one in B, and it shares each of its three outside links with a different one of the other three plaquettes; by (a), any two plaquettes of Y share exactly one link. Label the links of q cyclically $m_B, \ell_1, \ell_2, \ell_3$ with $m_B \in B$ (so ℓ_1 and ℓ_3 are opposite), let q_i be the plaquette of $Y \setminus \{q\}$ containing ℓ_i, and let $v := \ell_1 \cap \ell_2$, $v' := \ell_2 \cap \ell_3$. The plaquettes q_1, q_2 both contain v and share a link m; by (b), $v \in m$, since otherwise $q_1 = q_2$. Likewise q_2, q_3 share a link $m' \ni v'$. As q_1 shares only ℓ_1 with q, $m \neq \ell_2$, and likewise $m' \neq \ell_2$. So q_2 contains ℓ_2 and a further link at each endpoint of ℓ_2: it is spanned by ℓ_2 and a unit vector $e \perp \ell_2$, with m the link from v in direction e and m' the link from v' in direction e. The vector e is not parallel to ℓ_1: if $m = \ell_1$, then $q_2 \ni \ell_1$, but q_2 shares only ℓ_2 with q; and if m and ℓ_1 point in opposite directions from v, they are collinear and no plaquette contains both, whereas q_1 does. Hence e is perpendicular to ℓ_1 and ℓ_2, q_1 is the plaquette spanned at v by ℓ_1 and e, and q_3 the plaquette spanned at v' by ℓ_3 and e. Since $\ell_3 = \ell_1 + (v' - v)$, $q_3 = q_1 + (v' - v)$ is the translate of q_1 by a unit vector perpendicular to its plane, so q_1 and q_3 have no common vertex and share no link: a contradiction. Hence $|Y| \geq 5$. □

The bound is sharp: the five faces other than p_0 of a unit cube containing p_0 satisfy the hypotheses with $B = E(p_0)$.

Lemma 5.5 (Two-point bounds in finite volume)**.** *Let $0 < \beta < \beta_0(d)$ and let Λ be periodic with all sides ≥ 3. Put*

$$q_1 := eK\, e^{(K+1)\theta}\epsilon \;\leq\; x < 1,$$

$$\mathcal{T}_Q := (1+\epsilon)^6 e^{(24d-42)\theta}\Big(1+\frac{1}{K}\Big)^{24(d-2)}\frac{q_1^5}{1-q_1}, \qquad \mathcal{T}_p := (1+\epsilon)e^{(8d-11)\theta}\Big(1+\frac{1}{K}\Big)^{K}\frac{q_1^5}{1-q_1},$$

$$\bar u := \frac{(1+\iota)\beta+\beta^2 e^{\beta}}{2N^2}, \qquad a_* := N\bar u(1+\mathcal{T}_p)+\sqrt{\tau_0}\,\mathcal{T}_p, \qquad \eta := \frac{\epsilon}{\sqrt{\tau_0}}+\frac{a_* N\bar u}{\tau_0},$$

$$L(\beta) := \frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\,\frac{N^4}{\tau_0^2}\,\mathcal{T}_Q,$$

$$V(\beta) := (1+\epsilon)(1+\mathcal{T}_p)+\mathcal{T}_p,$$

with ϵ, x, θ, K as in the Convention. If $\eta \leq 1$, then $a := \langle A\rangle_{\Lambda,\beta}$ satisfies $|a| \leq a_*$, and

$$\mathrm{Cov}_{\Lambda,\beta}(A, A_1) \;\geq\; \tau_0\, u(\beta)^4\, L(\beta), \qquad \mathrm{Var}_{\Lambda,\beta}(A) \;\leq\; \tau_0\, V(\beta).$$

Proof. The expansion. For $S \subseteq E(\Lambda)$ and a bounded measurable H depending only on the links of S, the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] applies verbatim (with E_C replaced by S, as in Definition 2.3) and gives

$$\langle H\rangle_{\Lambda,\beta} = \sum_{X\in\mathcal{X}_S}\Phi_H(X)\,r_X, \qquad \Phi_H(X) := \int H\prod_{p\in X} f_\beta(U_p)\,dU, \qquad r_X := \frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}.$$

If a link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_H(X) = 0$ ([III, Lemma `al:lem:facts`](c)). Always $|\Phi_H(X)| \leq \epsilon^{|X|}\int|H|\,dU$, by $\|f_\beta\|_\infty \leq \epsilon$.

Let $(S,\mathcal{C})$ be $(S_Q,\mathcal{Q})$ or $(S_A,\{p_0\})$. Every plaquette not in $\mathcal{C}$ contains at most one link of S: two links of S in one plaquette either meet at a corner of Q (resp. p_0), and are then two perpendicular edges which span a face of Q (resp. p_0 itself), or are opposite links of the plaquette, i.e. parallel edges whose base points differ by a unit vector, which (sides ≥ 3) are edges of a common face. Each link lies in $2(d-1)$ plaquettes ([III, Lemma `al:lem:facts`](d)), and each edge of Q (resp. p_0) in two faces of Q (resp. in p_0); hence $|\mathcal{P}_{S_Q}| = 6+12(2d-4) = 24d-42$, $|\mathcal{P}_{S_Q}\setminus\mathcal{Q}| = 24(d-2)$, $|\mathcal{P}_{S_A}| = 1+4(2d-3) = 8d-11$ and $|\mathcal{P}_{S_A}\setminus\{p_0\}| = K$.

For $X\subseteq\mathcal{C}$ we have $E(X)\subseteq S$, hence $\overline{X} = \mathcal{P}_S$ and $r_X = r_\emptyset =: R_S$; and $\sum_{X\subseteq\mathcal{C}}\Phi_H(X) = \int H\prod_{p\in\mathcal{C}}(1+f_\beta(U_p))\,dU =: M_S(H)$. Let $X\in\mathcal{X}_S$ with $X\not\subseteq\mathcal{C}$ and $\Phi_H(X)\neq 0$, and put $Y := X\setminus\mathcal{C}$. A link of $E(Y)\setminus S$ lies in at least two plaquettes of X, none of which is in $\mathcal{C}$ (whose plaquettes have all their links in S); so Lemma 5.4 applies with $B = S$ and gives $|Y|\geq 5$. Every component of Y contains a plaquette of $\mathcal{P}_S\setminus\mathcal{C}$: it lies in a component of X, which contains a plaquette of $\mathcal{P}_S$; if the component of Y is not itself a component of X, one of its plaquettes shares a link with a plaquette of $X\cap\mathcal{C}$, a link of S. Hence, by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] applied to the graph on $(\mathcal{P}\setminus\mathcal{C})\cup\{o\}$ whose root o is joined to the J' plaquettes of $\mathcal{P}_S\setminus\mathcal{C}$, there are at most $\binom{J'+Kn}{n} \leq (1+\frac{1}{K})^{J'}(eK)^n$ possible Y with $|Y| = n$. Moreover $\overline{X} = \mathcal{P}_S\cup\{p : E(p)\cap E(Y)\neq\emptyset\}$, so $|\overline{X}| \leq |\mathcal{P}_S|+(K+1)|Y|$ and, by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)], $r_X \leq e^{\theta|\mathcal{P}_S|+\theta(K+1)|Y|}$. Summing over $X\cap\mathcal{C}$ (at most $(1+\epsilon)^{|\mathcal{C}|}$ in total weight) and over Y,

$$\big|\langle H\rangle_{\Lambda,\beta} - R_S\,M_S(H)\big| \;\leq\; \int|H|\,dU\cdot(1+\epsilon)^{|\mathcal{C}|}e^{\theta|\mathcal{P}_S|}\Big(1+\frac{1}{K}\Big)^{J'}\sum_{n\geq 5}q_1^n, \tag{1}$$

which is $\int|H|\cdot\mathcal{T}_Q$ for $(S_Q,\mathcal{Q})$ and $\int|H|\cdot\mathcal{T}_p$ for $(S_A,\{p_0\})$. (Here $q_1\leq x$ because $(K+1)\theta < 1$, [III, Lemma `al:lem:constants`].)

The plaquette p_0. With $H = 1$: $M_{S_A}(1) = \int(1+f_\beta) = 1$, so $|R_{S_A}-1| \leq \mathcal{T}_p$. With $H = A$: $\int A(1+f_\beta(U_{p_0}))\,dU = \mathrm{Re}\int\chi_\rho w_\beta\,dU/c_0 = c_{\bar\rho}/c_0 = Nu$ (as $\int\chi_\rho\chi_r = \delta_{r\bar\rho}$), and

$\int|A| \le (\int A^2)^{1/2} = \sqrt{\tau_0}$; so $|a| \le (1+\mathcal{T}_p)Nu + \sqrt{\tau_0}\,\mathcal{T}_p \le a_*$, by Lemma 5.1(iii). With $H = A^2$: $M_{S_A}(A^2) \le (1+\epsilon)\tau_0$, so $\mathrm{Var}(A) \le \langle A^2\rangle \le (1+\mathcal{T}_p)(1+\epsilon)\tau_0 + \tau_0\mathcal{T}_p = \tau_0 V$.

Positivity. By Lemma 5.2 with $B = \mathrm{Re}\,\chi_\rho$ evaluated on p_0 (so that $B(0) = A$, $B(1) = A_1$), $b = a$ and $E_0 = E(p_0)$,

$$\mathrm{Cov}(A, A_1) \;\ge\; \langle H_Q\rangle, \qquad H_Q := (A-a)(A_1-a)\prod_{s\in X_0}\varphi_\beta(U_s),$$

a function of the twelve edges of Q.

The cube. With $H = 1$: $M_{S_Q}(1) = \int\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\,dU = 1 + \int\prod_{p\in\mathcal{Q}} f_\beta(U_p)\,dU =: M_1$. (For $\emptyset \ne X \subsetneq \mathcal{Q}$, some edge of Q lies in exactly one face of X, so $\int\prod_X f_\beta = 0$.) By Lemma 5.1(ii) with $a_0 = a_1 = f_\beta$, $M_1 = 1 + \sum_{r\ne 0} d_r^2 u_r^6 \in [1, 1+\epsilon^6]$. So $R_{S_Q}M_1 \ge 1 - \mathcal{T}_Q$ by (1). With $H = H_Q$: since $\varphi_\beta(1+f_\beta) = f_\beta$,

$$M_Q := M_{S_Q}(H_Q) = \int \tilde a(U_{p_0})\,\tilde a(U_{p_1})\prod_{s\in X_0} f_\beta(U_s)\,dU, \qquad \tilde a := (\mathrm{Re}\,\chi_\rho - a)(1+f_\beta).$$

The function $\tilde a$ is a real class function invariant under inversion, and its character expansion converges absolutely (that of f_β does, and multiplication by $\mathrm{Re}\,\chi_\rho$ maps χ_r to a combination of characters with total dimension-weighted coefficient $\le Nd_r$). By Lemma 5.1(ii), $M_Q = \sum_{r\ne0}\hat a_r^2 u_r^4 \ge \frac{2}{1+\iota}\hat a_\rho^2 u^4$ (the terms $r = \rho, \bar\rho$, equal), where

$$\hat a_\rho = \int \tilde a\,\overline{\chi_\rho}\,dU = \tau_0 + \int \mathrm{Re}\,\chi_\rho\, f_\beta\,\overline{\chi_\rho}\,dU - a\,Nu \;\ge\; \tau_0 - \epsilon\sqrt{\tau_0} - a_*N\bar u = \tau_0(1-\eta) \ge 0,$$

using $\int \mathrm{Re}\,\chi_\rho\overline{\chi_\rho} = \tau_0$, $\int(1+f_\beta)\overline{\chi_\rho} = Nu$ and $|\int \mathrm{Re}\,\chi_\rho f_\beta\overline{\chi_\rho}| \le \epsilon(\int(\mathrm{Re}\,\chi_\rho)^2)^{1/2}(\int|\chi_\rho|^2)^{1/2}$. Since $\frac{2}{1+\iota}\tau_0^2 = \tau_0$, $M_Q \ge \tau_0(1-\eta)^2u^4$.

Next, $\int|H_Q|\,dU \le (\tau_0 + a^2)(\int|\varphi_\beta|)^4$. Indeed, write b_ℓ, t_ℓ for the bottom and top edges of Q over $\ell \in E(p_0)$ and v_y for its temporal edges. The holonomy of the side over $\ell = (y,i)$ is conjugate to $U_{b_\ell}U_{v_{y+e_i}}U_{t_\ell}^{-1}U_{v_y}^{-1}$ or its inverse, so for fixed bottom and temporal edges the map from $(U_{t_\ell})_\ell$ to $W := (U_{s_\ell})_\ell$ is a bijection of G^4 preserving Haar measure. By the Cauchy–Schwarz inequality, $\int|H_Q| \le (\int(A-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}(\int(A_1-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}$. In the first integral, A depends only on the bottom edges and W is Haar distributed independently of them, so it equals $\int(A-a)^2\cdot(\int|\varphi_\beta|)^4 = (\tau_0+a^2)(\int|\varphi_\beta|)^4$ by Lemma 5.1(i); the second is the same, with bottom and top exchanged. By Lemma 5.1(iv), $\int|H_Q| \le (\tau_0 + a_*^2)e^{16\beta}N^4u^4/\tau_0^2$.

Finally, by (1), $M_Q \ge 0$ and $R_{S_Q} \ge (1-\mathcal{T}_Q)/M_1 \ge (1-\mathcal{T}_Q)/(1+\epsilon^6)$,

$$\langle H_Q\rangle \;\ge\; R_{S_Q}M_Q - \mathcal{T}_Q\int|H_Q| \;\ge\; \tau_0u^4\Big[\frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\frac{N^4}{\tau_0^2}\mathcal{T}_Q\Big] = \tau_0u^4L(\beta).$$

□

Theorem 5.6 (Upper bound)**.** *Let $d \ge 3$, G metrisable, ρ irreducible and non-trivial, $0 < \beta < \beta_0(d)$, and with the notation of Lemma 5.5 suppose that $\eta(\beta) \le 1$ and $L(\beta) > 0$. Put $D(\beta) := \log(V(\beta)/L(\beta))$. Then*

(i) $m(\mu_\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$*;*

(ii) $m_{\Lambda_s}(\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$ *for every periodic spatial lattice Λ_s with all sides ≥ 3.*

As $\beta \downarrow 0$, $D(\beta) = 2(1+2/\sqrt{\tau_0})\beta + O(\beta^2)$; in particular the hypotheses hold for all sufficiently small β.

Proof. Write Γ_1 and Γ_0 for the covariance of A_1 and A and for the variance of A in the relevant theory. By Lemma 5.5, $\Gamma_1 \ge \tau_0u^4L > 0$ and $\Gamma_0 \le \tau_0V$, so $\log(\Gamma_0/\Gamma_1) \le 4\log(1/u) + D(\beta)$, provided these bounds hold for the theory in question. It remains to show $m \le \log(\Gamma_0/\Gamma_1)$.

(ii) Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \geq 3$, and A the corresponding gauge-invariant function on the spatial configurations, so that $A_1 = A(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. By [F, Theorem 'Ground-state expectations' (`thm:gs`)] (applied to $A_1 A$, to A and to A^2, using $\langle A(1)\rangle_\Lambda = \langle A(0)\rangle_\Lambda$ by translation invariance, [F, Theorem `thm:basic`](iv)), as $L_0 \to \infty$, $\mathrm{Cov}_\Lambda(A_1, A) \to \langle Q\hat{A}\Omega, e^{-H} Q\hat{A}\Omega\rangle$ and $\mathrm{Var}_\Lambda(A) \to \|Q\hat{A}\Omega\|^2$, where Q is the projection orthogonal to the vacuum, as in the proof of [F, Theorem 'Gap $\Leftrightarrow$ clustering in time' (`thm:gapclust`)](i); and $\langle Q\hat{A}\Omega, e^{-H} Q\hat{A}\Omega\rangle \leq e^{-m_{\Lambda_s}(\beta)} \|Q\hat{A}\Omega\|^2$. The bounds of Lemma 5.5 are uniform in L_0, so they hold for the limits Γ_1, Γ_0, and $e^{-m_{\Lambda_s}(\beta)} \geq \Gamma_1/\Gamma_0$.

(i) By [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) (note $\beta < \beta_0(d) < \beta_*(d)$, Lemma 4.2), the covariance and the variance under μ_β are limits of those on periodic lattices with all sides tending to infinity, so the bounds of Lemma 5.5 hold for them. With ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)] and $\tilde{A} := A - \nu(A)$, a function of the time-zero slice, [II, Corollary `iv:cor:chain`](iv) gives $\Gamma_1 = \langle \tilde{A}, P\tilde{A}\rangle_\nu$ and $\Gamma_0 = \|\tilde{A}\|_\nu^2$; since $\tilde{A} \perp \mathbf{1}$ and $\|P|_{\mathbf{1}^\perp}\| = e^{-m(\mu_\beta)}$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)], $\Gamma_1 \leq e^{-m(\mu_\beta)}\Gamma_0$.

Finally, as $\beta \downarrow 0$: $\epsilon = 2\beta + O(\beta^2)$, $x, \theta, q_1 = O(\beta)$, so $\mathcal{T}_Q, \mathcal{T}_p = O(\beta^5)$; $\bar{u}, a_* = O(\beta)$ and $\eta = 2\beta/\sqrt{\tau_0} + O(\beta^2)$. Hence $\log V = 2\beta + O(\beta^2)$ and $\log L = -2\eta + O(\beta^2)$. □

Lemma 5.7 (The constants on the whole range)**.** *Let $d \geq 3$ and $0 < \beta < \beta_0(d)$. Then $\epsilon < 8.06 \cdot 10^{-4}$, $\mathcal{T}_Q < 8.2 \cdot 10^{-4}$, $\mathcal{T}_p < 1.5 \cdot 10^{-5}$, $\eta < 1.15 \cdot 10^{-3}$, $V(\beta) < 1.00085$ and*

$$L(\beta) \;>\; 0.9969 - 8.2 \cdot 10^{-4}\, \frac{N^4}{\tau_0^2}.$$

In particular, if $N \leq 3$ (e.g. $G = U(1)$, $SU(2)$, $SU(3)$), then $L(\beta) > 0.73$ and $D(\beta) < 0.32$ for all $0 < \beta < \beta_0(d)$.

Proof. $\beta_0(d)$ decreases with d, and $\beta < \beta_0(d)$ means $\epsilon < \epsilon_0(d) := \frac{1}{e^2 K(K+2)} \leq \frac{1}{168 e^2} < 8.056 \cdot 10^{-4}$. By [III, Lemma `al:lem:constants`], $x < \frac{1}{K+2}$ and $\theta < \frac{1}{K+1}$; so $q_1 \leq x < \frac{1}{K+2}$ and $\frac{q_1^5}{1-q_1} < \frac{1}{(K+2)^4(K+1)}$. With $K = 8d - 12 \geq 12$: $(24d-42)\theta < \frac{24d-42}{8d-11} < 3$, $(8d-11)\theta < 1$, $(1+\frac{1}{K})^{24(d-2)} = (1+\frac{1}{K})^{3K-12} < e^3$ and $(1+\frac{1}{K})^K < e$. Hence $\mathcal{T}_Q < 1.00485\, e^6/(14^4 \cdot 13) < 8.12 \cdot 10^{-4}$ and $\mathcal{T}_p < 1.00081\, e^2/(14^4 \cdot 13) < 1.49 \cdot 10^{-5}$. Next $\beta < \beta_0(3) < 4.027 \cdot 10^{-4}$ gives $N\bar{u} < 4.03 \cdot 10^{-4}/N$, $a_* < 4.18 \cdot 10^{-4}$ (as $\sqrt{\tau_0} \leq 1$), and, with $\tau_0 \geq \frac{1}{2}$, $\eta < \sqrt{2} \cdot 8.056 \cdot 10^{-4} + 2 \cdot 4.18 \cdot 10^{-4} \cdot 4.03 \cdot 10^{-4} < 1.14 \cdot 10^{-3}$. So $(1-\eta)^2(1-\mathcal{T}_Q)/(1+\epsilon^6) > 0.99691$, $(1 + a_*^2/\tau_0) e^{16\beta} < 1.00647$, which gives the bound on L; and $V < 1.000806 \cdot 1.0000149 + 0.0000149 < 1.00085$. For $N \leq 3$, $N^4/\tau_0^2 \leq 4 \cdot 81 = 324$, so $L > 0.9969 - 0.2657 > 0.73$ and $D < \log(1.00085/0.73) < 0.32$. □

Corollary 5.8 (Two-sided bounds on the whole range)**.** *Let $d \geq 3$, G metrisable and ρ irreducible and non-trivial of dimension $N \leq 3$. For every $0 < \beta < \beta_0(d)$,*

$$4 \log \frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{u(\beta)} + D(\beta), \qquad 0 < D(\beta) < 0.32, \quad D(\beta) = O(\beta);$$

explicitly,

$$4 \log \frac{1}{\beta} - 4 \log(4e^3 K) - 4 \log \frac{e^{4\beta} - 1}{4\beta} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{\beta} + 4 \log \frac{2N^2}{1+\iota} + 4\beta + D(\beta).$$

The same bounds hold for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 4. For every N the upper bounds hold for all sufficiently small β (Theorem 5.6); in particular $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$, and the same holds uniformly for the finite-volume gaps on spatial lattices with all sides ≥ 4.

Proof. The lower bounds are Theorem 1.2(ii) and (i), with $q(\beta) = 4e^3 K\beta \cdot \frac{e^{4\beta}-1}{4\beta}$. The upper bounds are Theorem 5.6 and Lemma 5.7 (which also gives $\eta \le 1$), with $\log \frac{1}{u} \le \log \frac{2N^2}{(1+\iota)\beta} + \beta$ (Lemma 5.1(iii)). $D > 0$ because $V \ge 1 \ge L$. □

Remark 5.9 (Numbers, the first version, and checks). (i) Numerically (`python3` with `mpmath`), the largest value of $D(\beta)$ on $(0, \beta_0(d))$, attained as $\beta \uparrow \beta_0(d)$, is $4.8 \cdot 10^{-3}$, $4.9 \cdot 10^{-2}$ and $3.8 \cdot 10^{-3}$ for $SU(2)$, $SU(3)$ and $U(1)$ in $d = 3$, and $1.4 \cdot 10^{-3}$, $1.0 \cdot 10^{-2}$ and $1.3 \cdot 10^{-3}$ in $d = 4$; and $D(\beta)/\beta \to 6$ for $SU(2)$, $\to 2(1 + 2\sqrt{2}) \approx 7.66$ when $\iota = 0$. So on the whole range the upper bound is $4\log(1/u)$ up to a few per cent. The two-sided window of Corollary 5.8 still has width about $4\log(4e^3K) + 4\log\frac{2N^2}{1+\iota}$ (about 35.1 for $SU(2)$ in $d = 4$); all of it comes from the constant of the lower bound (Remark 6.3(ii)).

(ii) The first version of this section (9 November 2026; see the repository history and `journal/2026-11-09b.md`) obtained the upper bound from the duplicated expansion of Section 2: the four sides of Q form the unique minimal connecting set, with the exactly computable term $\Phi_F(X_0) = 2\tau_0 u^4$, and every other connecting set has at least five plaquettes. There the remainder, with sup-norm weights $\approx 4\beta \cdot eK$ per plaquette, had to be compared with the leading term $\tau_0 u^4$, $u \approx (1+\iota)\beta/(2N^2)$, which restricted the result to $\beta \lesssim 10^{-16}$. Lemma 5.2 removes this comparison: the remainder in Lemma 5.5 is relative to the tube itself, and the only price is the factor $(\int |\varphi_\beta|/u)^4 \le e^{16\beta} N^4/\tau_0^2$.

(iii) Checks (a compass, not part of the proofs). Exact transfer-matrix computations for $\mathbb{Z}_2$ and $\mathbb{Z}_3$ in $d = 3$ on a 3×3 spatial torus (physical spaces of dimension 2^{10} and 3^{10}): all sixteen terms of the decomposition in the proof of Lemma 5.2 are non-negative, the covariance dominates $\langle H_Q\rangle$, $\langle H_Q\rangle/(\tau_0 u^4) \to 1$, and for $\beta < \beta_0(3)$ the bounds of Lemma 5.5 hold with room (e.g. $\langle H_Q\rangle/(\tau_0 u^4) = 0.9999997 \ge L = 0.998$ at $\beta = 4 \cdot 10^{-4}$ for $\mathbb{Z}_2$). On this lattice the true gap is $\approx 3\log(1/u)$ (a torelon around a side of length 3), so Theorem 5.6(ii) is valid but not sharp there, as Remark 6.3(iii) expects. For S_3 with its two-dimensional representation, on a spatial graph consisting of one square, the decomposition was checked against brute-force summation over gauge-fixed configurations. Lemma 5.4 (with $B = E(Q)$ and $B = E(p_0)$) was checked by exhaustive enumeration on the tori $(3,3,3)$ and $(4,4,4)$ up to five plaquettes (the only five-plaquette sets are the caps: the five faces other than F of a unit cube $\ne Q$ sharing a face F with Q, resp. of a unit cube containing p_0) and on $(3,3,3,3)$ and $(4,3,3,3)$ up to four plaquettes.

6 Two dimensions; discussion

Proposition 6.1 (Two dimensions)**.** *Let $d = 2$ and $0 < \beta < \beta_0(2)$, so that $K = 4$ and $q(\beta) < 0.909$ (Lemma 4.2).*

(i) For every $L_1 \ge 3$, the spatial lattice $\Lambda_s = \mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta) \ge L_1 \log(1/q(\beta))$.

(ii) In infinite volume, $\mathrm{Cov}_{\mu_\beta}(A, B) = 0$ for all A, B as in Theorem 1.2(iii) (time ranges with $a_2 < b_1$); hence $m(\mu_\beta) = \infty$.

Proof. By Lemma 3.1, $g = L_1$, and Theorem 4.1 gives the bound with $q^{L_1\tau}$. (i) follows as in the proof of Theorem 1.2(i), with L_1 in place of 4. For (ii), the bound for the periodised observables on tori with $L_0, L_1 \to \infty$ tends to 0, since $\tau \ge 1$ and $g = L_1 \to \infty$, and the torus covariances converge to $\mathrm{Cov}_{\mu_\beta}(A, B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2) < \beta_*(2)$, Lemma 4.2); then [II, Theorem `iv:thm:OS`](iii) applies with every $m > 0$. □

Remark 6.2 (Consistency with the exact solution). In $d = 2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_r c_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the

exact finite-volume gap is $L_1 \log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 6.1 reproduces both features (with q in place of u). It is also consistent with [II, Proposition 'Gauge invariance of the ground-state chain' (`iv:prop:phys`)]: in $d = 2$ the spatial links of a time slice form a line, gauge-invariant functions of finitely many of them are constant, and P acts trivially on the physical sector. The difference between $d = 2$ and $d \geq 3$ is the girth of the spatial graph (L_1 versus 4): in $d \geq 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 6.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta) \sim 4\log(1/u(\beta))$. Theorem 1.2 gives the lower half, $m \geq 4\log(1/\beta) - O(1)$ for all $\beta < \beta_0(d)$, and Theorem 5.6 the upper half, $m \leq 4\log(1/u(\beta)) + O(\beta)$, on the same range when $N \leq 3$ and for sufficiently small β in general; Corollary 5.8 combines them. The upper bound uses a single correlator at time separations 0 and 1; by the positivity of P ([II, Remark 'Consequences for the numerics' (`iv:rem:numerics`)]) any larger pair of separations could be used instead. The additive constants of the two halves differ ($-4\log(4e^3K)$ versus $+4\log\frac{2N^2}{1+\iota}$); the strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$, i.e. the upper one. (ii) The constant $-4\log(4e^3K)$ reflects the crude counting of [III] and the crude bound $\|g_p\|_\infty \leq e^{4\beta} - 1$: for $SU(N)$ and $U(1)$, $c_0 = 1 + O(\beta^2)$ gives $\|g_p\|_\infty \leq 2\beta + O(\beta^2)$, which improves the constant by $4\log 2$, and counting only surfaces without free edges, as in [III, Remark 'Constants, sharpness, improvements' (`al:rem:constants`)](ii), would improve it further. (iii) The hypothesis $L_i \geq 4$ in (i) is needed: if a spatial side equals 3, Theorem 4.1 holds with $g = 3$, and closed strips of three plaquettes per layer around that side (torelons, carrying the flux of a Polyakov loop) are expected (heuristic) to give states of energy $\approx 3\log(1/u(\beta))$, below $4\log(1/q(\beta))$ for very small β; exact computations for $\mathbb{Z}_2$ and $\mathbb{Z}_3$ on a 3×3 spatial torus show exactly this (Remark 5.9(iii)). (iv) Nothing here is non-abelian: the argument applies verbatim to $G = U(1)$, whose $d = 4$ theory is expected to be massless at large β (proved in [FS82] for the Villain action and asserted there for Wilson's; [III, Remark 'The $U(1)$ test' (`al:rem:u1test`)]). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is controlled with the right leading behaviour (Corollary 5.8), which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have $\geq g$ temporal plaquettes per crossed layer ($g = 4$ for $d \geq 3$, spatial sides ≥ 4)	Lem. 3.1, 3.2, 3.3, Cor. 3.4	Theorem
Covariance bound $q^{g\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m(\mu_\beta) \geq 4\log(1/q(\beta)) = 4\log(1/\beta) - O(1)$, and $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$ for spatial sides ≥ 4; $d \geq 3$, $\beta < \beta_0(d)$	Thm. 1.2	Theorem
$d = 2$: gap $\geq L_1 \log(1/q)$; infinite gap in infinite volume	Prop. 6.1	Theorem
Positivity: $\mathrm{Cov}(A(0), A(1))$ dominates the expectation in which the four sides of the cube carry φ_β (every β)	Lem. 5.2	Theorem
Cube identity $\sum_{r\neq 0} \hat{a}_{0,r}\hat{a}_{1,r}u_r^4$; five-plaquette lemma	Lem. 5.1, 5.4	Theorem
Matching upper bound $m \leq 4\log(1/u(\beta)) + D(\beta)$, $D = O(\beta)$ (infinite volume; finite volume, sides ≥ 3): for all $\beta < \beta_0(d)$ if $N \leq 3$ ($D < 0.32$), for small β in general	Thm. 5.6, Lem. 5.7	Theorem
$4\log(1/q) \leq m(\mu_\beta) \leq 4\log(1/u) + D$ for all $\beta < \beta_0(d)$ ($N \leq 3$; also finite volume, sides ≥ 4); $m = 4\log(1/\beta) + O(1)$	Cor. 5.8	Theorem
A lower bound with the constant of the upper one, $m \geq 4\log(1/u) - O(\beta)$	Rem. 6.3(ii)	open

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